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Predicting and optimising the energy yield of perovskite-on-silicon tandem solar cells under real world conditions†

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ET model
motivation

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Metal halide perovskite absorber materials have emerged as a potential new technology for large-scale low-cost photovoltaic solar power. One great advantage lies in the ability to tune their light absorption band across the visible to near infrared spectral regions, making it an ideal candidate for tandem solar cell applications, in combination with traditional crystalline silicon. For a multi-junction solar cell to operate at peak efficiency, the current generation in all junctions must closely match, especially for monolithically integrated tandem architectures. It is feasible to achieve such matching under a standardized solar spectrum with direct illumination. However, under real world conditions the spectrum of sun light, and the fraction of diffuse to direct sun light varies considerably depending upon the location and weather conditions. Hence, it is not directly obvious how much more efficient a multi-junction solar cell needs to be, in comparison to a single junction cell, before it will produce more electrical power under real world conditions. Here, we introduce a rigorous optical and device simulation to optimize perovskite-on-silicon tandem solar cells and identify feasibility of various optimisation parameters to achieve the highest possible efficiencies. Firstly, we determine that the ideal bandgap for a perovskite "top-cell" is 1.65 eV, which will deliver up to 32% efficiency when combined with a silicon rear cell. Furthermore, we calculate the annual energy yield under hourly spectrum changes at different locations and optimize the stack to show that tandem solar cells are yielding up to 30% more energy output than the single junction silicon. Most critically, the standardized air mass 1.5 efficiency measurement improvements observed for the tandems cells, translate almost entirely to the same fractional improvement in energy yield. Hence, the efficiency of the tandem cell is not significantly "de-rated" by real world spectral variations. We do observe however, that tandem solar cell stacks can deliver further improvements by optimising differently depending on the location of installation. Our results justify the drive towards monolithically integrated multi-junction solar cells, and will enable guidance to design the ideal perovskite tandem device and allow estimations for energy yield and hence the levelized cost of electricity.

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Broader context

Perovskite solar cells have quickly become one of the new hopes in the renewable energy sector due to their potential to lead to highly efficient and low-cost solar power conversion. While the efficiency of single absorber layer devices has already reached over 22%, a successful market entry is likely to require even higher values to compete with existing technologies, and offset the comparative risk of deploying a new PV technology. The remarkable ability to tune the absorbing region of these materials, allows them to be combined with a complimentary absorbing silicon solar cell to form so-called tandem solar cells. However, doubts arise whether such devices are only performing theoretically better under standardised testing conditions or if they will also yield more energy in real world conditions. We used a combination of optical and electrical simulations together with annually measured weather data. We determine that with optimised materials and layer thicknesses, depending on the location and the sun tracking system, over 30% high efficiencies will translate to equally high gains in energy output. Our study confirms and justifies a drive towards perovskite-on-silicon tandem cells, to deliver a step increase in energy yield utility scale photovoltaic power generation.

Introduction

Hybrid organic–inorganic metal halide perovskites have optical absorption characteristics that not only make them very interesting for single layer solar cell applications but also for tandem devices.

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With a bandgap greater than 1.5 eV, the absorption spectrum of typical hybrid metal-halide perovskites is very strong in the ultraviolet and visible light region but transparent for infrared (IR) light, which makes it an ideal top absorbing layer to combine with materials that absorb well in the longer wavelength section of the spectrum.[1] Crystalline silicon as well as copper-indium gallium selenide (CIGS), have been identified and already demonstrated[2–8] as feasible candidates to couple with perovskites in tandem solar cells. Combined power conversion efficiencies of up to 23.7%[9] in a monolithic 2-terminal (2T) configuration[5] and 25.2% in a 4-terminal (4T) configuration[8] on top of silicon have been achieved experimentally and are expected to continue to increase rapidly over the next few years. Specifically perovskite-on-silicon tandem modules are strong candidates to be the first commercial embodiments of perovskite solar cells.[10] The cost benefit of increasing the power output per installed photovoltaic module is very attractive, since the installation cost of solar panels usually accounts for a substantial part of the total system cost.[11] Since most of the non-module costs of a solar installation scale with the area of the installation, significant cost savings are made by boosting the power output of each module. The potentially inexpensive manufacturing methods of perovskite solar cells integrated into silicon cell processing make it an attractive premise, since the added marginal cost should be compensated within a short time by increased energy production.

The use of perovskite solar absorbers as top cells has become even more interesting since it has been demonstrated that the bandgap can be precisely changed to higher values by adding different halide contents to the MA or FA based perovskite crystals[12,13] and the use of caesium to stabilize the crystalline perovskite phase.[14–17] A wider bandgap allows for the perfect optimisation of a tandem solar cell to reach the maximum achievable efficiency, as we show in Fig. S6 (ESI†). Since the viability of bandgap shifts to lower energies have been shown by using Sn–Pb based perovskites,[18,19] this has led to the further development of perovskite-on-perovskite tandem solar cells, in addition to ''hybrid'' tandem solar cells, with efficiencies crossing 20%.[20,21]

Engineering a tandem device stack proves to be challenging, due to the complexity of adding new material layers that need to be electronically and optically optimized. Monolithically integrated stacks (also referred to as two terminal devices), that connect both solar cells in series, need to have well designed recombination layers. Whereas four-terminal (4T) devices need to have two transparent electrodes upon either side of each junction, in order to transmit the longer wavelength region of the light through the front cell into the rear cell, and conduct current out of each cell. The additional layers introduce more interference effects that can lead to reduced absorption in the photovoltaic absorber layers, and require fine-tuning to achieve maximum performance.

Lately there have been several reports undertaking optical modelling to optimize the thicknesses of each layer in perovskite-on-silicon tandem solar cell stacks. Filipic *et al.*[22] compared in a model, based on the transfer matrix method, the maximum efficiencies achievable with 2T *versus* 4T tandem devices. They found that efficiencies above 30% are theoretically feasible and that 4T devices show a higher maximum power output. However, the authors did not take into account the optical losses induced by the physical separation of the top and the bottom cell as well as the fact that there will arise additional efficiency losses and illumination area losses due to cell interconnection and charge extraction from the internal electrodes.

Further optical simulation results included the comparison of different bandgap perovskite materials. This has been achieved by theoretically shifting the published optical constants of the perovskite along the energy axis of the spectrum,[23] as well as by deriving experimentally the optical constants of different perovskite material compositions *via* fitting experimental reflection and transmission data.[24]

A critical question arises when developing tandem solar cells: if a tandem solar cell is more efficient in a standard efficiency measurement, will it quantifiably output more energy under real world operating conditions? Solar cell efficiency is determined by measuring the power output of a solar cell under a standard solar spectrum, generally with collimated light. The typical standard spectrum is termed air mass (AM) 1.5, and is equivalent to the direct sun light passing through 1.5 (perpendicular) atmospheres on a clear day, with 100 mW cm^{-2} irradiance. However, the precise spectral composition of the real solar spectrum varies depending upon many factors such as atmospheric dust, humidity and cloud cover. In addition, on a cloudy or hazy day, a significant fraction of the sun light is diffuse, which has a different spectrum from the direct sun light and will hit the solar panel at a range of angles. A monolithic multi-junction solar cell relies on having equal amounts of light absorption in all sub cells in order to maximize efficiency. However, with varying spectra and incident angles of the solar irradiance, the sub cells will usually be out of the current-matching regime. Therefore, comparing the lab-measured efficiency of a tandem solar cell with the efficiency of a single junction solar cell is of only limited value, unless we know how the improved efficiency relates to increased energy yield under real world solar illumination. The effect that such spectral fluctuations can have on the performance of amorphous and microcrystalline silicon tandem devices has been studied before and shows variations of up to 20% over the same time period under different weather conditions.[25,26] Slightly less drastic results of about 4% variation have been shown for polycrystalline solar cells.[27] In addition to estimating the energy yield of the tandem solar cells, it may be useful to optimize the device stacks differently in different locations, in order to maximize the energy yield from the multi-junction solar cells.

EY model motivation

Energy yield (EY) simulations have been carried out in more and less complex ways for several photovoltaic materials and tandem device architectures.[28–30] A common approach to include varying spectra into the calculations is done *via* binning the spectra by their average photon energy (APE).[31,32] Futscher and Ehrler[33] have recently carried out such a calculation, in which they modelled the spectral influence on perovskite-on-silicon tandem devices by using the combined detailed balance limit of the absorbers. It was found that specifically 4T, but also 2T architectures are not significantly affected by the spectral variations

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and would perform relatively as well as the single junction silicon solar cells. A simulation with more rigorous device modelling was carried out for CdTe/CdS tandem applications,[29] which lead to the results that 4T configuration are more desirable due to less dependence on the varying spectrum by current matching conditions.

Here, we present the implementation of a rigorous optical and device model, which we use to simulate perovskite-on-silicon tandem solar cells, in combination with real weather data and complex diffuse light treatment. We use this to predict the annual energy yield at specified northern hemisphere locations with three different sun-tracking systems, fixed axis (facing south with fixed tilt angle), 1-axis (facing south with fixed tilt angle and east-west rotating axis) and 2-axis tracking. In addition to estimating the energy yield for cells optimized for maximising the AM1.5 solar cell efficiency, we also present a fast optimising algorithm, which allows us to tune the device stack in order to maximize the energy yield for a specific location. Consequently, we determine further improvements in energy yield of up to 1.4%, *via* bespoke optimisation of the device stack. Furthermore, we compared the energy yields of perovskite-on-silicon tandem solar cells as well as single junction silicon and perovskite solar cells in relation to their AM1.5 irradiance power conversion efficiencies, and introduce an "efficiency de-rating value". We find that tandem devices not only show higher power conversion efficiencies (PCEs) but that this gain translates almost equally into energy yield under real world conditions, and the 2T tandem devices are usually within 1% of their expected energy yield. This means that the expected enhancement in energy yield of up to 30% in tandems over single junction silicon cells should be realized. We also demonstrate that 2T configurations come very close to 4T performances, without accounting for additional optical and electrical losses that might arise in such 4T architectures. Our results and simulation methods present a roadmap for the development and evaluation of optimized high-energy-yield tandem devices that will benefit research and the design strategies for cell manufacturing. Our findings also justify a significant focus on advancing perovskite-on-silicon tandem solar cells.

Results and discussion

We describe the details of the optical and electronic modelling in the ESI.† In brief, we took the optical constant of each material in the device stack from literature,[22] and undertook a transfer matrix method in order to calculate the fraction of light absorbed in each layer in the device stack. For our electrical model, we used a single diode equivalent circuit, including the radiative efficiency of the solar cell, in addition to ideality factor, shunt and series resistance, and extracted the parameters from published current voltage curves on real measured cells. We could subsequently simulate *J–V* characteristic curves and find the maximum power point. We describe our device model in detail in the ESI.† We modelled three different types of devices, as we illustrate in Fig. 1a. The monolithically connected 2T stack consists of a perovskite top cell and a silicon heterojunction with intrinsic thin layer (SHJ) bottom cell, which was changed only slightly for the 4T configuration by virtually electrically detaching the two devices at the indium tin oxide (ITO) layer. The layers, starting from the top illumination side, are MgF_2 as an anti-reflective coating, ITO, 2,2′,7,7′-tetrakis-(*N*,*N*-di-4-methoxyphenylamino)-9,9′-spirobifluorene (Spiro-OMeTAD), perovskite (varied bandgaps), electron transporting layer (varied between TiO_2, C_{60} and SnO_2), ITO, and then the SHJ cell consisting of 10 nm p-type amorphous silicon, 5 nm intrinsic amorphous silicon, 3.5 mm of intrinsic single crystalline silicon, 5 nm intrinsic amorphous silicon and 10 nm of n-type amorphous silicon, with an additional bottom electrode of 300 nm Ag. We note that similar to other reports,[22] the thickness of the single crystalline silicon layer of 3.5 mm is much larger than the usual wafer thickness of 200 μm, however, this emulates a commonly used efficient back reflector, which increases the optical path length of light travelled through the silicon absorber. We analyse the single junction SHJ cell separately with the only difference that

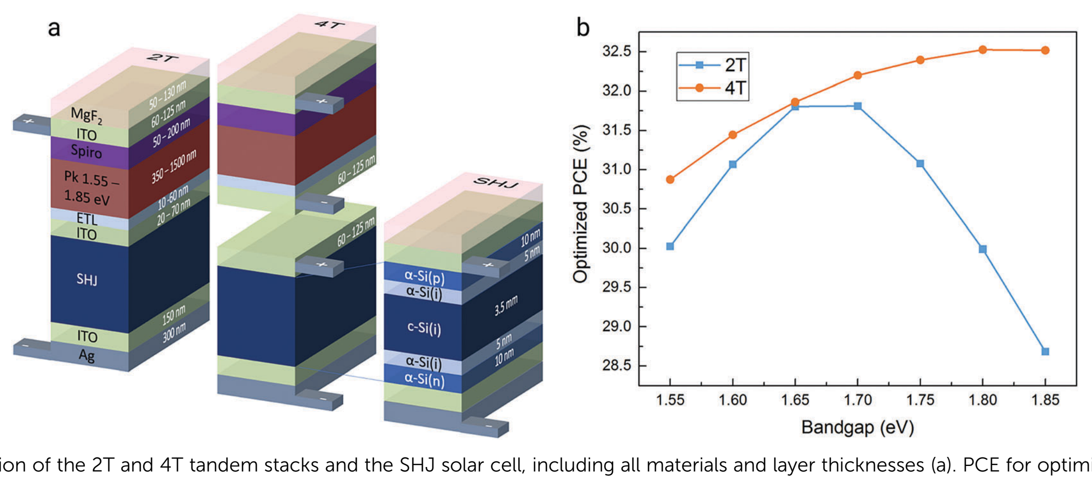

Fig. 1 Illustration of the 2T and 4T tandem stacks and the SHJ solar cell, including all materials and layer thicknesses (a). PCE for optimized 2T and 4T tandem device stacks with C_{60} as the electron-transport layer for shifted perovskite bandgaps (b).

we included a top layer of MgF_2 to minimize reflection losses. We set the upper and lower limits of the range of thicknesses and bandgaps, which we allowed to vary (which we also show in Fig. 1a) to guarantee feasibility of electrical functionality and manufacturability.

Initially, as a quality check, we carried out a simple device performance simulation of a single junction perovskite solar cell resembling the stack reported by Saliba *et al.*[34] We determined the current density–voltage (*JV*) fitting parameters (found in the ESI†) for our device model *via* fitting the diode parameters to match the experimental current–voltage data of one of the highest efficiencies published for a fabricated perovskite solar cell. We present the simulated EQE spectrum and *JV* characteristic in Fig. S7a and b (ESI†). Our results are sufficiently close to the reported experimental values, which allows us to confidently progress with the more complex modelling of the multi-junction device stacks. The simulated perovskite solar cell reached a power conversion efficiency (PCE) of 21.8%, which is identical to the experimentally achieved efficiency. We carried out this calculation with an AM1.5 spectrum and 0° incidence angle (which we refer to as AM1.5/0° condition) to resemble lab conditions. Throughout the remainder of the electrical modelling, we assume the same shunt- and series resistance, ideality factor and radiative efficiency for all perovskite cells with varying bandgap. Hence, our calculated efficiencies for cells of differing bandgap assume that equally high quality perovskite solar cells can be fabricated over the relevant range of bandgaps. We plot the modelled *JV* curves and their characteristic performance parameters for different light intensities and varying bandgaps in Fig. S7a–e (ESI†).

We took a similar approach for the simulation of the single junction SHJ cell. However, since reflection losses play a more important role for silicon solar cells, we modelled the device with an anti-reflection layer of MgF_2, and varied the thicknesses of the top two layers (ITO and MgF_2) through a differential evolution algorithm to maximize the PCE. We find that thicknesses of 71 nm MgF_2 and 63 nm ITO lead to the highest PCE of 22.7%. We present the simulated EQE spectrum and *JV* curve in Fig. S8 (ESI†), which resemble well the shapes and values shown by others.[22] For the Si cell, we employed the diode characteristics of the real silicon cell reported in ref. 35. Here, we note that significantly higher J_{SC}s have been measured for silicon cells by using physically textured anti-reflective coatings. However, including texture into the optical model is more computationally expensive and we have not done so here.

In a next step, we vary the bandgap of the perovskite material. Since experimental data for a complete range of perovskite bandgaps are not available, we shift the absorption coefficient along the energy axis and apply a Kramers–Kronig transformation to calculate the complex refractive index, as we explain in the ESI.† We present the results of the original and shifted optical constants in Fig. S9 (ESI†). The shifted extinction coefficients appear similar in shape to the published experimental data of absorbance spectra for different perovskite bandgaps.[14] We then optimize the 2T and 4T perovskite-on-silicon device stacks in order to maximize the power conversion efficiency for AM1.5/0° irradiance, with the optical constants that correspond to bandgaps of the perovskite cells ranging from 1.55 eV to 1.85 eV. We present the bandgap energies and the resulting maximum PCEs in Fig. 1b. We discover that the optimal top cell bandgap for the perovskite cell in the 2T configurations, peaks between 1.65 and 1.70 eV. This is slightly lower than previously published reports[2,14] that took Shockley–Queisser models (which we calculated and plotted in Fig. S6, ESI†) for their reason to aim for 1.75 eV as the ideal bandgap for the top absorber when coated onto a silicon absorber with a bandgap around 1.1 eV. Furthermore, as others have also observed, the perovskite bandgap shift in the 4T configuration has less impact, and the combined PCE for the 4T tandem reaches a maximum with a perovskite bandgap at a slightly higher energy of 1.8 eV. The crystalline silicon cells still possess fundamentally fewer electronic losses than the perovskite cells, which is quantifiable by the difference in energy between the absorber bandgap and the open-circuit voltage. Therefore, when optimising for 4T tandem efficiency, without the requirement for current matching, more weighting is given to the silicon rear cell and the optimum bandgap for the perovskite top cell moves to higher energy. For both, 2T and 4T architectures, PCEs over 31% are feasible with realistically constrained diode characteristics, and the 4T device appears to perform slightly better, pushing beyond 32%.

Another pertinent question is how much of an effect different interface materials have on the overall performance of tandem devices, with respect to their impact upon the light absorption and optical optimisation. Therefore, we compared the three different materials C_{60}, to TiO_2 and SnO_2 as electron transporting layers (ETL) between the perovskite and the ITO interfaces. Again, we performed the same optimisation procedure for AM1.5/0° conditions and two different perovskite bandgaps (1.55 eV and 1.70 eV). We show the results in Fig. S11 (ESI†), where we observe a small effect of the ETL on the maximum PCE and can therefore be almost neglected for optical performance considerations. We retain C_{60} as the ETL material for all subsequent simulations.

Now that we have found the optimal device stack and perovskite bandgap to maximize the PCE of the 2T device, we plot the simulated EQE spectrum (Fig. 2a) and *JV* characteristic curve (Fig. 2b) together with the tabulated optimized thicknesses (inset). We list the individual device performance characteristics in Table 1. The varying fringes are noticeable in the EQE spectrum in the red region, which are a result of interference stemming from reflected light within the many thin layers. Introducing scattering layers to break the coherency of the light would improve this behaviour, which implies that further efficiency gains, beyond what we have calculated here, should be feasible through experimental optimisations. Moreover, it can be seen from the *JV* curves in Fig. 2b that the currents from the absorbing layers are very well matched, leading to the optimal device efficiency. We have had to maximize the perovskite layer thickness, almost to its upper limit of 1.2 μm, to match the relatively high current (around 20 mA cm^{-2}) that comes from the transmitted photons absorbed by the silicon cell. Since most currently published perovskite cells have a thickness of less than 1 micrometre, this already indicates that increasing the absorber

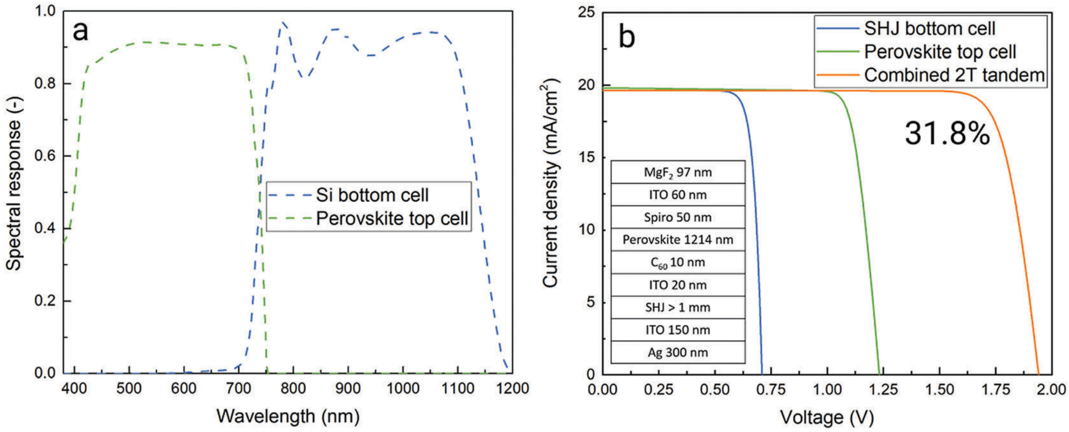

Fig. 2 Simulated EQE spectrum (a) and *JV* characteristic with performance parameters (b) of optimized 2T silicon-on-perovskite (1.65 eV) tandem cell. Table of optimized layer thicknesses (inset).

Table 1 *JV* curve performance parameters of individual sub cells and combined 2T tandem cell for AM1.5/0° optimized devices

	PCE (%)	FF (−)	V_{oc} (V)	J_{SC} (mA cm^{-2})
SHJ (unfiltered)	22.82	0.830	0.74	37.21
SHJ bottom cell	11.60	0.834	0.71	19.65
Perovskite 1.65 eV	20.25	0.832	1.23	19.82
Combined 2T	31.81	0.834	1.94	19.65

thickness is an important direction for future optimisation. The fact that the highest PCE can be achieved with a 1.65 eV perovskite bandgap is another promising detail, since the currently best performing single-junction perovskite solar cells in literature were reported to have only slightly lower bandgaps of 1.63 eV.[34] Notably, the optimum bandgap for a top-cell to combine with silicon determined by SQ modelling, approximately 1.75 eV, significantly deviates from what we estimate here, where we have employed realistic material and device parameters.

The analysis of optimized tandem stacks under AM1.5/0° conditions is interesting from a research lab perspective because these are the standard conditions, which can be ubiquitously used to experimentally characterize solar cells. However, for real world applications, which should be the long-term focus for the development of solar cells, the annual energy yield at a particular deployment location has a lot more significance for financial and engineering feasibility analysis. We selected three locations in the USA with significantly different weather conditions for comparison, and included fixed, 1- and 2-axis solar tracking. We give the detailed location and weather information for the analysed locations of Golden (Colorado), Seattle (Washington) and Mohave desert (California) in Table S1 and Fig. S12 (ESI†). The solar irradiation characteristic, the precipitation and the temperature vary largely at all locations. As expected, the Mohave desert is exposed to a lot more direct beam irradiation than diffuse light, caused by fewer clouds and therefore less scattered light. The lowest solar irradiation is received in Seattle, which is predictable due to the regularly overcast skies and the higher latitude. The change from the desert spectrum to the more overcast location's spectra is visible in some reduced irradiance in the red and infrared regions, which is due to absorption by water vapor in clouds. We chose these locations for the simulations since they offer a broad range of illumination conditions.

Next, we would like to understand and illustrate how much the tandem solar cell performance is influenced by changes in the spectral composition of the sun light, as compared to the single junction silicon cell. In order to do this, we chose two spectra, one corresponding to a typical summer midday spectrum in Mohave desert, which contains more photons in the IR and red region and one corresponding to a spring day in Golden which has a relatively enhanced blue component due to Raleigh scattering of particles in the air. We show the normalized photon flux of the spectra in Fig. 3a, and the non-normalized photon flux spectra for the entire spectral region in Fig. S13 (ESI†). Since the tandem cell is optimized for AM1.5/0°, both the IR and UV rich spectra result in current mismatch between the top and bottom cells in the 2T tandem device. However, in the electrical model, the FF of the *JV* curve is considerably enhanced when current mismatch occurs, and this largely compensates for the drop of photocurrent due to current mismatch. We show the *JV* curves for the AM1.5/0° optimized 2T tandem and the SHJ devices operating in the IR weak and IR rich spectra in Fig. 3b and c respectively. We list the corresponding performance parameters in Table 2. Surprisingly, for the IR weak spectrum the PCE of the 2T tandem cell is still maintained at 30.9% (a relative drop of 2.7% in comparison to AM1.5) despite the substantial loss of current (*ca.* 1.8 mA cm^{-2}) from the SHJ rear cell. Notably, in comparison, the efficiency of the single junction silicon heterojunction cell also decreases relatively by a similar amount of 2.6%, in the IR weak spectrum. This is because there are more thermalisation losses in the silicon cell under the IR weak spectrum. Under IR rich illumination, the PCE of the 2T tandem cell drops from 31.8% (under AM1.5) to 27.0%, whereas the single junction silicon cell only decreases in efficiency from 22.7 to 22.1% under the same spectrum. The large efficiency loss of the 2T tandem device in this scenario is due to a significant decrease in the

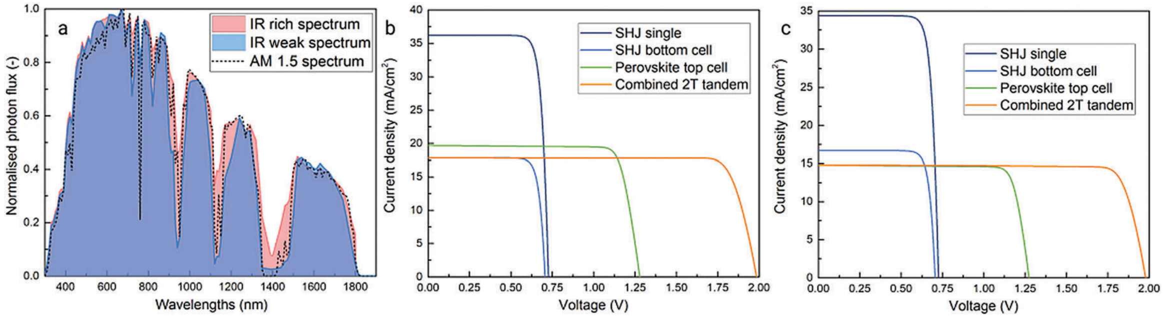

Fig. 3 Comparison of normalized photon flux of IR rich and IR weak solar spectra in comparison to AM1.5 spectrum (a). Modelled *JV* curves of 2T tandem and SHJ devices under (b) IR weak (c) and IR rich spectra for perovskite bandgap of 1.7 eV.

Table 2 Tabulated *JV* curve characteristic parameters of optimized 2T tandem and SHJ devices under IR weak and IR rich spectra. As compared to the AM1.5 spectrum the demonstrated spectra have integrated powers below 100 mW cm^{-2}, as we note in the right column

	PCE (%)	FF (–)	V_{oc} (V)	J_{SC} (mA cm^{-2})	Spectral power (mW cm^{-2})
2T combined IR weak	30.93	0.860	1.98	17.85	98.53
SHJ IR weak	22.09	0.826	0.73	36.21	‖
SHJ (bottom) IR weak	10.66	0.833	0.71	17.85	‖
Perovskite (top 1.7 eV) IR weak	21.33	0.835	1.28	19.69	‖
4T combined IR weak	31.99	—	—	—	‖
2T combined IR rich	26.98	0.860	1.98	14.81	93.36
SHJ IR rich	22.10	0.829	0.72	34.38	‖
SHJ (bottom) IR rich	10.50	0.832	0.71	16.69	‖
Perovskite (top 1.7 eV) IR rich	17.07	0.845	1.27	14.81	‖
4T combined IR rich	27.57	—	—	—	‖

above perovskite bandgap photons, leading to missed current in the perovskite cell. We note that these presented spectra have lower integrated power than 100 mW cm^{-2} of the AM1.5 spectrum. Hence, the SHJ cell PCE decreases is in part due to the reduced illumination power, which decreases the V_{oc}. We expected the 2T tandem solar cell efficiency to vary with spectral irradiance. However, it is also apparent that the single junction silicon cell varies under differing illumination conditions. Therefore, it is not obvious which cell architecture will fair comparatively better or worse under real world variable illumination conditions. Notably, under the differing spectrum we have shown here, the absolute PCE of the 2T tandem cell remains considerably higher than the single junction silicon cell.

In order to allow us to quantify the real world changes to the spectral distribution of direct and diffuse light in comparison to the standardized AM1.5 spectrum, we calculate the average of all direct normal irradiance (DNI) and diffuse horizontal irradiance (DHI) spectra, weighted by the integrated photon flux, and subsequently normalized to an integral of 1. We plot the results for the different locations together with the normalized AM1.5 photon flux in Fig. S14 (ESI†) and then integrate over the blue to visible absorption region of perovskites (280–760 nm) as well as the IR absorption region of silicon (760–1200 nm) to get the respective integrated photon flux for each spectral regime. We tabulate the findings in Table S2 (ESI†) together with the relative differences to the corresponding values of the AM1.5 spectrum. Our calculations quantify that the more overcast a location, the lower the visible fraction and the higher is the infrared fraction of the direct normal irradiance compared to the AM1.5 spectrum. Precisely the opposite is true for locations with less cloud cover, since we determine blue shifted diffuse irradiation and slightly red shifted direct irradiation in the Mohave desert.

The second component influencing the optical model is the incident angle of the photons. Since in real world conditions, the position of the sun towards the solar panel is permanently changing, it will have an important influence on the overall energy yield. Due to geometric considerations, the short-circuit current-density of a solar cell with perfect angular dependence should drop with angle of incidence θ as the cosine of θ. We simulate the angular dependent performance of the SHJ cell and the AM1.5/0° optimized 2T tandem cell, and plot the J_{SC} normalized together with the cosine dependence of the intensity in Fig. S15a (ESI†). Although both cells fall slightly short of the cos(θ) curve, it is almost impossible to see the difference in performance between the two cells. Therefore, we present the relative difference of normalized J_{SC}s in Fig. S15b (ESI†). Encouragingly, we observe that the steeper the angle of incidence, the better the improvement of tandem J_{SC} and hence we would expect the 2T tandem cell to operate comparably better under diffuse light or when fixed to one tilt angle only.

Analogous to Fig. 1b, we carried out optimisations to see which bandgap material would yield the highest annual energy output at the three specified U.S. locations. We show the results for all different tracking systems and tandem architectures in Fig. 4a–c. Interestingly we find a slightly different bandgap dependence when optimising 2T tandems for energy yield as when optimising for PCE at AM1.5/0° conditions. We observe that in Seattle the peak in performance appears to be shifted towards a lower bandgap of the perovskite cell, which can be explained by the relative red-shift of the solar spectrum. This leads to an improvement of energy yield form 476 kW h m^{-2} to 481 kW h m^{-2} (*ca.* 1% gain). Also, the tracking seems to make a slight difference to the optimum bandgap, which can be observed in Fig. 4a, since better sun tracking in Golden leads to higher energy yields at a slightly lower bandgap of 1.65 eV instead of 1.7 eV. We can explain this with the blue shift of diffuse light due to Raleigh scattering from gas particles in the atmosphere as compared to the direct irradiation, which is less influenced by this effect (visible in Fig. S14 and Table S2, ESI†). Generally, we achieve notably higher energy with single and 2-axis tracking, which point the cell towards the high intensity direct irradiance beam. The fixed 4T tandem devices appear to have a peaking bandgap at higher energies. We plot the highest energy yields for each optimized device stack in the diagram in Fig. 4d, which demonstrates perfectly the efficacy of solar tracking systems and the importance of PV deployment location. We estimate energy yields of over 1000 kW h m^{-2} with a 4T tandem device, deployed in Mohave desert with a 2-axis tracker, which is reduced by almost 30% if no solar tracking is employed. Almost exactly half of this energy yield is to be expected from a similar device also 2-axis tracked in Seattle. The important findings are that 1-axis tracking systems are performing nearly as well as 2-axis trackers, especially for locations with more direct spectral distribution. Furthermore, 4T devices are only yielding a couple of percent more energy in all scenarios, as compared to the 2T tandems. We note that 4T architectures might suffer from additional optical and electronic losses due to the requirement of current extraction from the intermediate electrodes, which we have not accounted for in this comparison. We note that we have not included temperature dependencies in our energy yield estimations. Although increased operating temperature will certainly reduce the overall energy yield, towards the end of our manuscript, we will return to this in more detail and justify why we do not expect this to cause significant differences between the relative energy yields of the different cell types.

For completeness, we also calculated the energy yield for a 1.65 eV perovskite single junction cell, which exhibited an AM1.5/0° PCE of 22% and find that perovskite single junction devices yield in most locations almost as much energy as SHJ single junction devices. We show the energy yield for these perovskite single junction devices in Fig. 4d.

We now proceed to optimize the tandem device stacks for maximising the annual energy yield at different locations, as opposed to simply optimising the device stacks for AM1.5/0°. We present the results in Fig. 4e as relative annual energy yield gains for location optimized devices, in comparison with the devices optimized for AM1.5/0°. The EY gains that we determine show that relative improvements of up to 1.4% are possible. This is an important finding, since it indicates that device stacks for future commercial tandem applications would slightly benefit from optimisation for every deployment in different locations. However, we note that the largest gains are made with fixed axis deployment, and single and 2 axis tracking brings the variance down to below a 0.2% relative gain *via* location optimisation. We present the annual energy yields and the respective optimisation gains of all location optimized devices with different bandgaps in Fig. S17 and S18 (ESI†). Additionally, we list the layer thicknesses of the optimized 2T tandem devices with a perovskite bandgap of 1.65 eV for all locations and tracking systems in Table S3 (ESI†). We find that the perovskite layer thicknesses need to be increased when more IR irradiation is to be expected and when the devices are tracked, due to a shorter light path through the absorbing layer when the photons arrive in a normal incidence angle.

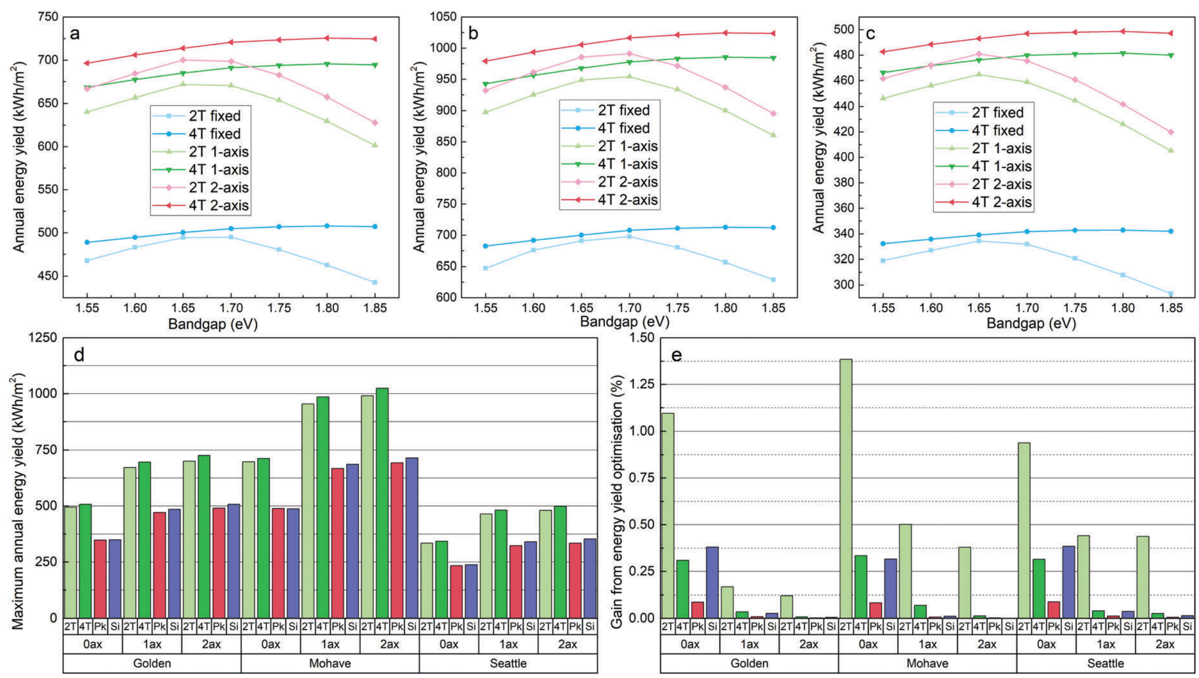

Fig. 4 PCE and energy yield of optimized 2T and 4T tandem solar cells depending on perovskite bandgap, optimized and calculated with fixed, 1-axis and 2-axis tracking at locations Golden (a), Mohave desert (b) and Seattle (c). Highest achievable annual energy yield for optimized devices with varying tracking system at different locations, including 2T and 4T perovskite-on-silicon tandem devices and SHJ (Si) and perovskite (Pk) single junction devices (d). Highest relative energy yield gain of devices optimized for energy yield compared to devices optimized for $AM_{1.5}/0°$ PCE (e).

The energy yield calculations are required in order to estimate the cost of installed PV power in different locations. To assess how well a photovoltaic product is performing in the real world in relation to its manufacturer's rated efficiency, it is common practice in industry to evaluate the so called annual capacity factor for a particular location, which is defined as the annual energy yield (in kW h) divided by the system's peak power rating (in W_p):

$$f_C\ (\mathrm{kW\,h/W_p}) = \frac{EY\ (\mathrm{kW\,h\,m^{-2}})}{PCE\ (\%) \times P_{\mathrm{sun}}\ (\mathrm{W\,m^{-2}})}$$

Additionally, for the development of the multi-junction technologies it would be useful to have a simple "correction factor" which could be applied to the AM1.5/0° PCE measurement to be able to make a like-to-like comparison between the perovskite-on-silicon tandem and single junction silicon cells. We therefore define an "efficiency de-rating factor" f_D as follows:

$$f_D = \frac{f_{C,\mathrm{tandem}}}{f_{C,\mathrm{SHJ}}}$$

Hence, in order to estimate how much of a real improvement in energy a relative increase in efficiency will deliver *via* making the tandem solar cells, the measured AM1.5/0° PCE of the tandem device should be multiplied by f_D. If $f_D > 1$, the tandem cell will perform even better than expected from the efficiency gain, if $f_D < 1$, the tandem cell with perform less-well than expected from the efficiency gain. We present our calculated de-rating factors for all device architectures, tracking systems and for the three different locations in Fig. 5. The outcome is very encouraging for the future prospects of perovskite-on-silicon tandem cells, since the de-rating factor is close to unity for most locations, ranging from 0.99 to 1.025. Therefore, on average applying a de-rating factor of $\sim$0.99 appears sensible and conservative, which implies that a 30% efficient tandem cell should be considered to be as effective as a $\sim$29.6% efficient single junction cell. We plot the de-rating factors of all tandem devices with different bandgaps in Fig. S18 (ESI†). Furthermore, we list the simulation results for EY, EY gain, capacity factors and derating factors in Table S4 (ESI†). We note that we estimate the highest de-rating factors (less de-rating) for untracked devices, which we assign to the better performance of tandem cells at steeper incidence angles. The efficiency of devices deployed in Seattle must be generally de-rated more, which arises from comparatively better power conversion of SHJ devices in red-shifted spectra. Moreover, we find that perovskite single junction devices perform remarkably well, specifically at untracked deployment and in clear skies. We can explain these results with the previously discussed spectral power distribution in red and visible regions, as stated in Table S2 (ESI†). Without tracking, the diffuse irradiation becomes more important, which is heavily blue shifted in comparison to the AM1.5 spectrum and therefore the perovskite cell is performing well in comparison to its AM1.5/0° rated PCE. However, in Seattle the direct irradiation spectrum is shifted more to the red and hence the silicon absorber performs relatively better.

A key factor, which we have omitted from our modelling, is the change in cell efficiency with temperature, and the variation of operational temperature of the solar cells under different locations (ambient temperature) and irradiance conditions. Solar cells usually reduce in efficiency with increasing temperature,[36,37] and due to thermal heat gain, the operational temperature of the solar cell will be above the ambient temperature. In order to estimate absolutely realistic energy yields, temperature should therefore be accounted for. However, in order to estimate the difference between perovskite-on-silicon tandem cells and SHJ single junction cells, we simply need to be reassured that the temperature coefficient of the cells will be similar. Using our combined optical and electronic model, we present modelling results of the temperature dependence of the different cells in the ESI.† We show that the temperature coefficient of perovskite solar cells is no worse than the temperature coefficient of silicon cells, as has been previously determined.[38] Additionally, we model the combined tandem temperature coefficient to be slightly lower than the one of SHJ cells, which supports our findings that the relative improvement in energy yield of tandem cells is expected, even when a full treatment, which accounts for temperature coefficients, is included. Moreover, others have assessed temperature effects in simple models and found variations in energy yield between silicon and perovskite-on-silicon tandem cells to be less than 1%.[39] We note that although modelling of the temperature dependence of the solar cell operation is relatively straight forward, knowing the temperature of the solar cell under every operating environment is very challenging. In order to do this we need to include a rigorous thermodynamic treatment accounting for the heat generation within the absorber material from thermalization of unutilized photon energy, combined with a heat conduction model through the surrounding materials (including lamination foils, glass, and frames) and the ambient, and a heat convection model. Even though ambient temperature and wind speed measurement data is available, a correct convection model requires accurate information about the specific racking system used for the installation and knowledge of all the thermal properties (emissivity, thermal conductivity, *etc.*) of all the layers employed in the photovoltaic module. This is beyond the scope of our current work, but will be the subject of a future development.

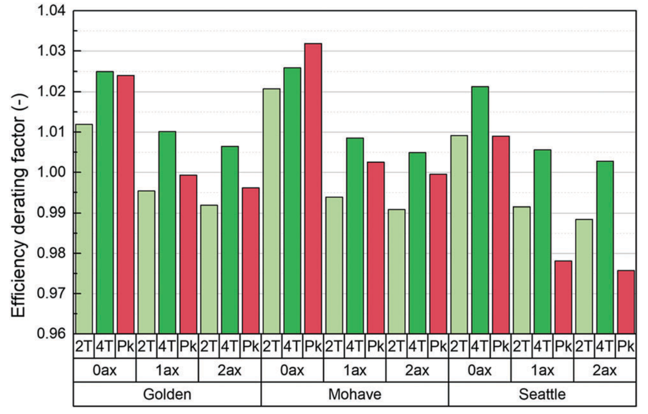

Fig. 5 Efficiency de-rating factors for best performing tandem devices with varying tracking systems at different locations.

One further omission from our model is the presence of physical texture, typically in the form of pyramids, which are ubiquitous on commercial silicon cells. The presence of physical texture both reduces reflection losses and minimizes angular dependence losses in real devices. The introduction of texture into our model would require an additional level of complexity, which is beyond the scope of our current study. However, we can assess the difference between single junction silicon and perovskite-on-silicon tandem cells in the limiting case where the cells exhibit no angular dependence, *i.e.* where the angular dependence of the photocurrent follows the cosine theta function. In order to do this, we artificially compress all the solar irradiation into the perpendicular direct normal and calculate the energy yield for the single and tandem junction cells and the de-rating factor. We plot the annual energy yield and efficiency de-rating factor results for the optimized devices in Fig. S19a and b (ESI†). We find that, even when disregarding angular dependence, the efficiency de-rating factors are not decreasing more than for tracked systems and remain within 1% of unity. Furthermore, we observe that in specific weather conditions, 2T tandem devices can even outperform single junction silicon solar devices ($f_D > 1$). We assume that this results from more thermalisation losses in a single junction SHJ cell under spectral conditions containing a higher concentration of blue irradiance.

Conclusion

In summary, we have developed a rigorous optical model and combined it with a solar cell device model in order to simulate the power conversion efficiency of perovskite-on-silicon tandem solar cells and estimate the energy yield for such solar cells deployed in differing environments. In addition, our optimisation routine has enabled us to determine the ideal material and thickness combinations which leads to the highest possible energy output, rather than simply maximising the AM1.5 measured power conversion efficiency. We arrived at surprising conclusions, that include the fact that perovskites should have a bandgap close to 1.65 eV in the 2T configurations and higher bandgaps of $\sim$1.80 eV in 4T designs. This implies that the currently best performing single junction perovskite solar cells that have been demonstrated in literature, have bandgaps close to this value and are therefore well suited for the integration in monolithic perovskite-on-silicon solar cells. Furthermore, we found that 2T tandems yield nearly as much energy as 4T designs and that 1-axis solar tracking systems deliver an additional comparable boost in energy yield. The deployment location plays a role, and we find slightly different device stacks deliver the highest energy yield in different locations. We have determined that in regions with IR richer spectra, lowering the bandgap of the perovskite top cell can lead to energy yield gains of 1%, when the panel is tracked. For fixed tilted deployment, the relative improvement by "customising" the tandem solar cell layer thicknesses for a specific location can be as much as 1.4%. However, for 1-axis tracking, the relative improvement upon customisation is less than 0.3%, indicating that a standard global design may be suitable for such installations. Most critically, we have assessed how the improvement in power conversion efficiency, going from single junction silicon to perovskite-on-silicon tandem cells, translates into improved energy yield. We find that, depending upon location and style of deployment (fixed or tracked), the perovskite-on-silicon cells will deliver between 98 and 103% of the expected energy yield. In addition, single junction perovskite cells could deliver substantially more energy than their silicon counterparts when deployed on fixed systems in clear sky locations. This implies that when making a side-by-side comparison between single junction silicon and perovskite-on-silicon tandem cells, the measured efficiency of the perovskite-on-silicon tandem cell is directly comparable to the efficiency of a single junction silicon cell. This gives good confidence that 2T perovskite-on-silicon tandem cells will deliver upon their promise of not just very high lab measured efficiencies, but also significantly boosted energy yields under real world conditions.

Conflicts of interest

Henry Snaith is founder and CSO of Oxford PV Ltd. a company commercialising perovskite photovoltaics.

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References

1 M. A. Green, A. Ho-Baillie and H. J. Snaith, The Emergence of Perovskite Solar Cells, *Nat. Photonics*, 2014, **8**, 506–514.
2 C. D. Bailie, M. G. Christoforo, J. P. Mailoa, A. R. Bowring, E. L. Unger, W. H. Nguyen, J. Burschka, N. Pellet, J. Z. Lee and M. Grätzel, *et al.*, Semi-Transparent Perovskite Solar Cells for Tandems with Silicon and CIGS, *Energy Environ. Sci.*, 2014, **8**, 956–963.
3 J. P. Mailoa, C. D. Bailie, E. C. Johlin, E. T. Hoke, A. J. Akey, W. H. Nguyen, M. D. McGehee and T. Buonassisi, A 2-Terminal Perovskite/silicon Multijunction Solar Cell Enabled by a Silicon Tunnel Junction, *Appl. Phys. Lett.*, 2015, **106**, 121105.
4 S. Albrecht, M. Saliba, J. P. Correa Baena, F. Lang, L. Kegelmann, M. Mews, L. Steier, A. Abate, J. Rappich and L. Korte, *et al.*, Monolithic Perovskite/silicon-Heterojunction Tandem Solar Cells Processed at Low Temperature, *Energy Environ. Sci.*, 2016, **9**, 81–88.
5 J. Werner, C.-H. Weng, A. Walter, L. Fesquet, J. P. Seif, S. De Wolf, B. Niesen and C. Ballif, Efficient Monolithic Perovskite/Silicon Tandem Solar Cell With Cell Area > 1 Cm 2, *J. Phys. Chem. Lett.*, 2016, **7**, 161–166.

6 B. Chen, Y. Bai, Z. Yu, T. Li, X. Zheng, Q. Dong, L. Shen, M. Boccard, A. Gruverman and Z. Holman, *et al.*, Efficient Semitransparent Perovskite Solar Cells for 23.0%-Efficiency Perovskite/Silicon Four-Terminal Tandem Cells, *Adv. Energy Mater.*, 2016, **6**, 1601128.

7 F. Lang, M. A. Gluba, S. Albrecht, J. Rappich, L. Korte, B. Rech and N. H. Nickel, Perovskite Solar Cells with Large-Area CVD-Graphene for Tandem Solar Cells, *J. Phys. Chem. Lett.*, 2015, **6**, 2745–2750.

8 J. Werner, L. Barraud, A. Walter, M. Bräuninger, F. Sahli, D. Sacchetto, N. Tétreault, B. Paviet-Salomon, S.-J. Moon and C. Allebé, *et al.*, Efficient Near-Infrared-Transparent Perovskite Solar Cells Enabling Direct Comparison of 4 – Terminal and Monolithic Perovskite/Silicon Tandem Cells, *ACS Energy Lett.*, 2016, **1**, 474–480.

9 K. A. Bush, A. F. Palmstrom, Z. J. Yu, M. Boccard, R. Cheacharoen, J. P. Mailoa, D. P. McMeekin, R. L. Z. Hoye, C. D. Bailie and T. Leijtens, *et al.*, 23.6%-Efficient Monolithic Perovskite/silicon Tandem Solar Cells with Improved Stability, *Nat. Energy*, 2017, **2**, 17009.

10 Oxford PV signs Joint Development Agreement with a major industrial partner | Oxford Photovoltaics http://www.oxfordpv.com/News/20161201-Oxford-PV-signs-Joint-Development-Agreement-with-a-major-industrial-partner, accessed Mar 2, 2017.

11 J. Jean, P. R. Brown, R. L. Jaffe, T. Buonassisi and V. Bulovic, Pathways for Solar Photovoltaics, *Energy Environ. Sci.*, 2015, **8**, 1200–1219.

12 G. E. Eperon, S. D. Stranks, C. Menelaou, M. B. Johnston, L. M. Herz and H. J. Snaith, Supplementary Information Formamidinium of Formamidinium Lead Trihalide: A Broadly Tunable Perovskite for Efficient Planar Heterojunction Solar Cells, *Energy Environ. Sci.*, 2014, **7**, 982.

13 J. H. Noh, S. H. Im, J. H. Heo, T. N. Mandal and S. Il. Seok, Chemical Management for Colorful, Efficient, and Stable Inorganic-Organic Hybrid Nanostructured Solar Cells, *Nano Lett.*, 2013, **13**, 1764–1769.

14 D. P. McMeekin, G. Sadoughi, W. Rehman, G. E. Eperon, M. Saliba, M. T. Horantner, A. Haghighirad, N. Sakai, L. Korte and B. Rech, *et al.*, A Mixed-Cation Lead Mixed-Halide Perovskite Absorber for Tandem Solar Cells, *Science*, 2016, **351**, 151–155.

15 M. Saliba, T. Matsui, J.-Y. Seo, K. Domanski, J.-P. Correa-Baena, M. K. Nazeeruddin, S. M. Zakeeruddin, W. Tress, A. Abate and A. Hagfeldt, *et al.*, Cesium-Containing Triple Cation Perovskite Solar Cells: Improved Stability, Reproducibility and High Efficiency, *Energy Environ. Sci.*, 2016, **9**, 1989–1997.

16 J. W. Lee, D. H. Kim, H. S. Kim, S. W. Seo, S. M. Cho and N. G. Park, Formamidinium and Cesium Hybridization for Photo- and Moisture-Stable Perovskite SolarCell, *Adv. Energy Mater.*, 2015, **5**, 1501310.

17 C. Yi, J. Luo, S. Meloni, A. Boziki, N. Ashari-Astani, C. Grätzel, S. M. Zakeeruddin, U. Röthlisberger and M. Grätzel, Entropic Stabilization of Mixed A-Cation ABX 3 Metal Halide Perovskites for High Performance Perovskite Solar Cells, *Energy Environ. Sci.*, 2016, **9**, 656–662.

18 F. Hao, C. C. Stoumpos, R. P. H. Chang and M. G. Kanatzidis, Anomalous Band Gap Behavior in Mixed Sn and Pb Perovskites Enables Broadening of Absorption Spectrum in Solar Cells, *J. Am. Chem. Soc.*, 2014, **136**, 8094–8099.

19 W. Liao, D. Zhao, Y. Yu, N. Shrestha, K. Ghimire, C. R. Grice, C. Wang, Y. Xiao, A. J. Cimaroli and R. J. Ellingson, *et al.*, Fabrication of Efficient Low-Bandgap Perovskite Solar Cells by Combining Formamidinium Tin Iodide with Methylammonium Lead Iodide, *J. Am. Chem. Soc.*, 2016, **138**, 12360–12363.

20 Z. Yang, A. Rajagopal, C. C. Chueh, S. B. Jo, B. Liu, T. Zhao and A. K. Y. Jen, Stable Low-Bandgap Pb–Sn Binary Perovskites for Tandem Solar Cells, *Adv. Mater.*, 2016, **28**, 8990–8997.

21 G. E. Eperon, T. Leijtens, K. A. Bush, R. Prasanna, T. Green, J. T.-W. Wang, D. P. McMeekin, G. Volonakis, R. L. Milot and R. May, *et al.*, Perovskite-Perovskite Tandem Photovoltaics with Optimized Band Gaps, *Science*, 2016, **354**, 861–865.

22 M. Filipic, P. Loper, B. Niesen, S. De Wolf, J. Krc, C. Ballif and M. Topic, $CH_3NH_3PbI_3$ Perovskite/Silicon Tandem Solar Cells: Characterization Based Optical Simulations, *Opt. Express*, 2015, **23**, A263–78.

23 S. Albrecht, M. Saliba, K. Jäger, L. Korte, A. Hagfeldt, M. Grätzel and B. Rech, Towards Optical Optimization of Planar Monolithic Perovskite/Silicon-Heterojunction Tandem Solar Cells, *J. Opt.*, 2016, **18**, 1–10.

24 M. Anaya, J. P. Correa-Baena, G. Lozano, M. Saliba, P. Anguita, B. Roose, A. Abate, U. Steiner, M. Grätzel and M. E. Calvo, *et al.*, Optical Analysis of $CH_3NH_3Sn_xPb_{1-x}I_3$ Absorbers: A Roadmap for Perovskite-on-Perovskite Tandem Solar Cells, *J. Mater. Chem. A*, 2016, **60–61**, 269.

25 T. Ishii, K. Otani, T. Takashima and Y. Xue, Solar Spectral Influence on the Performance of Photovoltaic (PV) Modules under Fine Weather and Cloudy Weather Conditions, *Prog. Photovoltaics Res. Appl.*, 2013, **21**, 481–489.

26 A. Virtuani and L. Fanni, Seasonal Power Fluctuations of Amorphous Silicon Thin-Film Solar Modules: Distinguishing between Different Contributions, *Prog. Photovoltaics Res. Appl.*, 2014, **22**, 208–217.

27 Y. Hirata and T. Tani, Output Variation of Photovoltaic Modules with Environmental Factors – I. The Effect of Spectral Solar Radiation on Photovoltaic Module Output, *Sol. Energy*, 1995, **55**, 463–468.

28 H. Liu, Z. Ren, Z. Liu, A. G. Aberle, T. Buonassisi and I. M. Peters, The Realistic Energy Yield Potential of GaAs-on-Si Tandem Solar Cells: A Theoretical Case Study, *Opt. Express*, 2015, **23**, A382.

29 J. P. Mailoa, M. Lee, I. M. Peters, T. Buonassisi, A. Panchula and D. N. Weiss, Energy-Yield Prediction for II–VI-Based Thin-Film Tandem Solar Cells, *Energy Environ. Sci.*, 2644, **9**, 2644–2653.

30 H. Liu, A. G. Aberle, T. Buonassisi and I. M. Peters, On the Methodology of Energy Yield Assessment for One-Sun Tandem Solar Cells, *Sol. Energy*, 2016, **135**, 598–604.

31 T. R. Betts, C. N. Jardine, R. Gottschalg, D. G. Infield and K. Lane, *Impact of Spectral Effects on the Electrical Parameters*

of Multijunction Amorphous Silicon Cells, In 3rd World Conference on Photovoltaic Energy Conversion, 2003, pp. 1–3.

32 S. R. Williams, T. R. Betts, T. Helf, R. Gottschalg, H. G. Beyer and D. G. Infield, *Modelling Long-Term Module Performance Based on Realistic Reporting Conditions with Consideration to Spectral Effects*, In Proceedings of the 3rd World Conference on Photovoltaic Energy Conversion, 2003, vol. B, pp. 1908–1911.

33 M. H. Futscher and B. Ehrler, Efficiency Limit of Perovskite/Si Tandem Solar Cells, *ACS Energy Lett.*, 2016, **1**, 2–7.

34 M. Saliba, T. Matsui, K. Domanski, J.-Y. Seo, A. Ummadisingu, S. M. Zakeeruddin, J.-P. Correa-Baena, W. R. Tress, A. Abate and A. Hagfeldt, *et al.*, Incorporation of Rubidium Cations into Perovskite Solar Cells Improves Photovoltaic Performance, *Science*, 2016, **354**, 206–209.

35 K. Masuko, M. Shigematsu, T. Hashiguchi, D. Fujishima, M. Kai, N. Yoshimura, T. Yamaguchi, Y. Ichihashi, T. Mishima and N. Matsubara, *et al.*, Achievement of More than 25% Conversion Efficiency with Crystalline Silicon Heterojunction Solar Cell, *IEEE J. Photovolt.*, 2014, **4**, 1433–1435.

36 M. A. Green, General Temperature Dependence of Solar Cell Performance and Implications for Device Modelling, *Prog. Photovoltaics Res. Appl.*, 2003, **11**, 333–340.

37 O. Dupré, R. Vaillon and M. A. Green, Physics of the Temperature Coefficients of Solar Cells, *Sol. Energy Mater. Sol. Cells*, 2015, **140**, 92–100.

38 S. N. Habisreutinger, T. Leijtens, G. E. Eperon, S. D. Stranks, R. J. Nicholas and H. J. Snaith, Carbon Nanotube/Polymer Composites as a Highly Stable Hole Collection Layer in Perovskite Solar Cells, *Nano Lett.*, 2014, **14**, 5561–5568.

39 B. C. Duck, R. B. Dunbar, O. Lee, K. F. Anderson, T. W. Jones, G. J. Wilson and C. J. Fell, *Energy Yield Potential of Perovskite-Silicon Tandem Devices*, In 2016 IEEE 43rd Photovoltaic Specialists Conference (PVSC), IEEE, 2016, pp. 1624–1629.